

SATELLITE DATA IN AGRICULTURAL AND ENVIRONMENTAL ECONOMICS

DAVID WUEPPER, HADI, WYCLIFE AGUMBA OLUOCH

Milan Summer School
16 – 18 September 2026

WELCOME

1

**New to Google
Earth Engine or R?**

2

**Already fluent in
either or both?**

3

**Few examples,
many uses.**

4

**Something
unclear? Ask.**

STATE OF AFFAIRS

- Earth's surface is changing
- Forests fall, cities sprawl, droughts spread, rivers shrink
- Satellites never look away
- They capture energy emitted or reflected off the surface
- But the energy is not information
- Pixel becomes knowledge when assigned a name/label
- Forest, cropland, water, built-up, solar panel, sand
- These are of interest to economists



Lon: 35.346455°

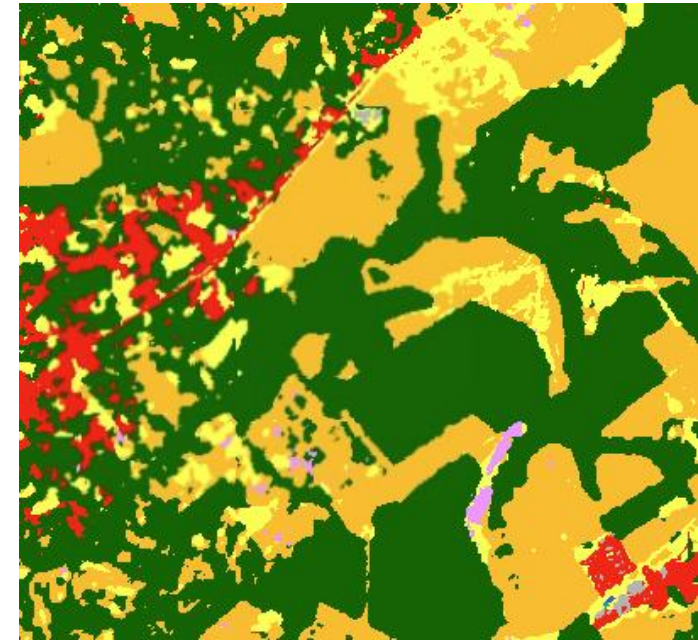
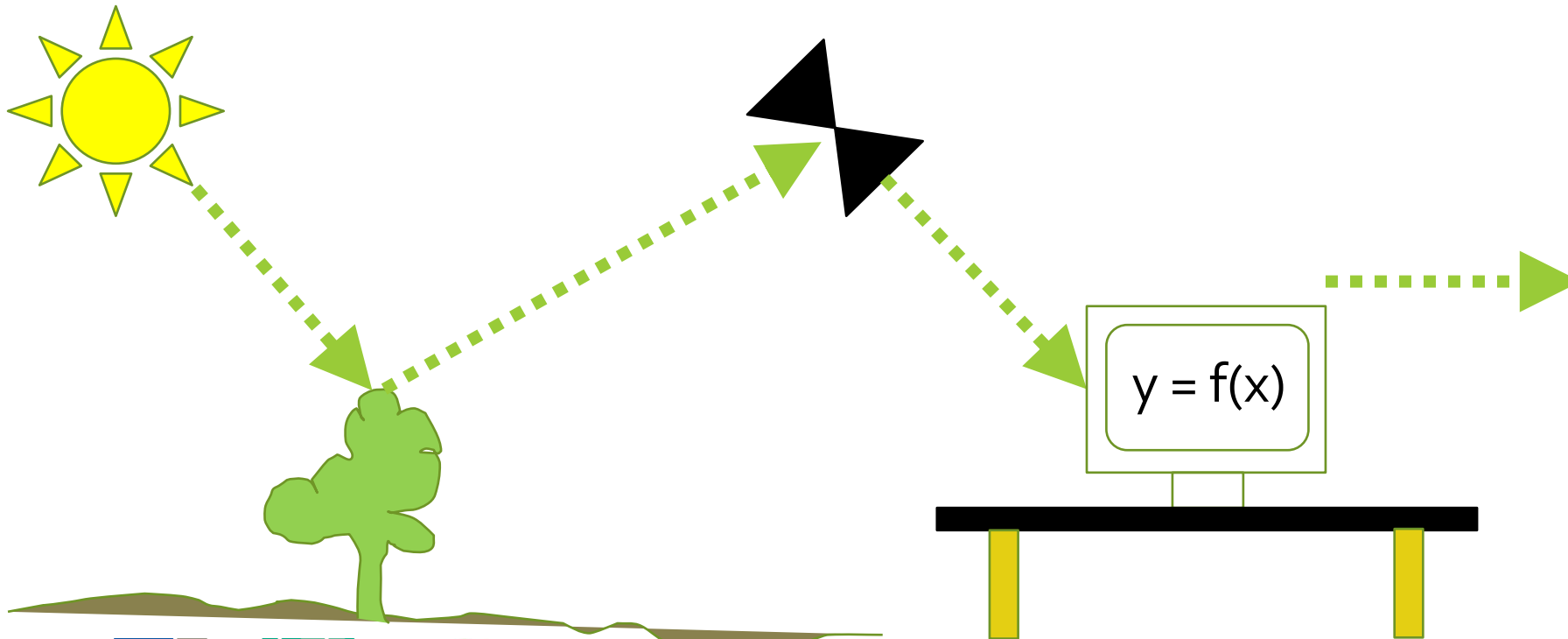


Lat: -0.372290°



THIS SESSION

Getting from energy to label, and knowing when that label can be trusted, is what this session is all about

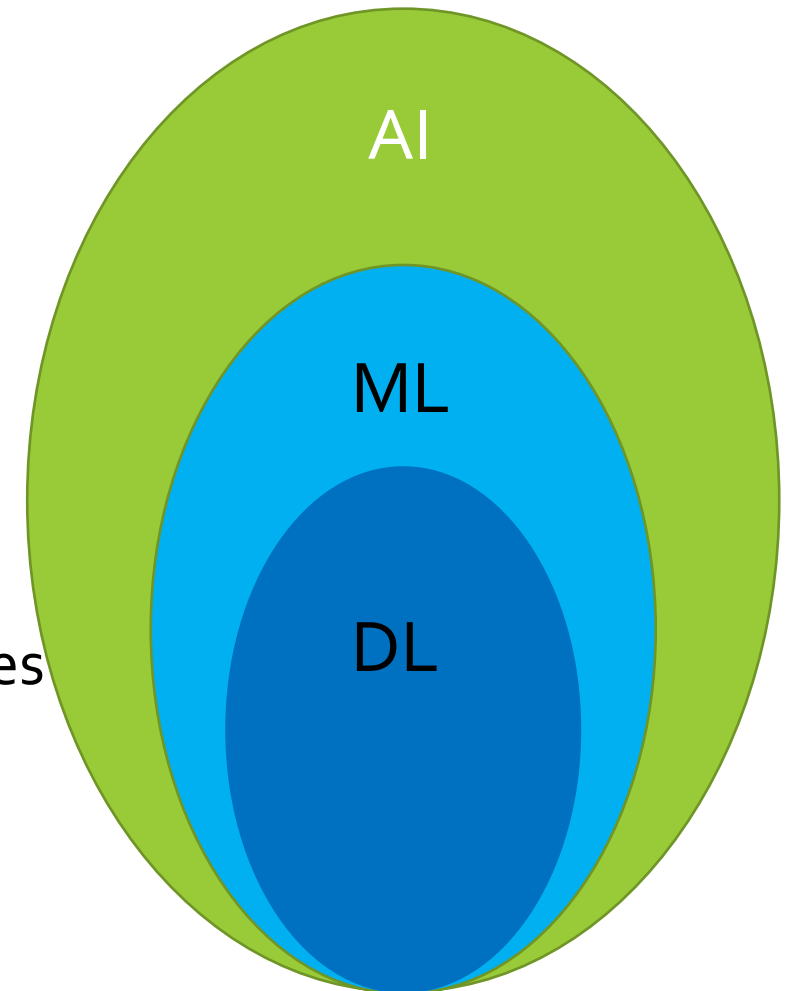


WHAT IS MACHINE LEARNING?

AI → When machines mimic **human-like intelligence**,
e.g: Crop disease detection from AI-powered drones

ML → When machines **learn from examples** to make
predictions,
e.g: Models predicting landcover from satellite images

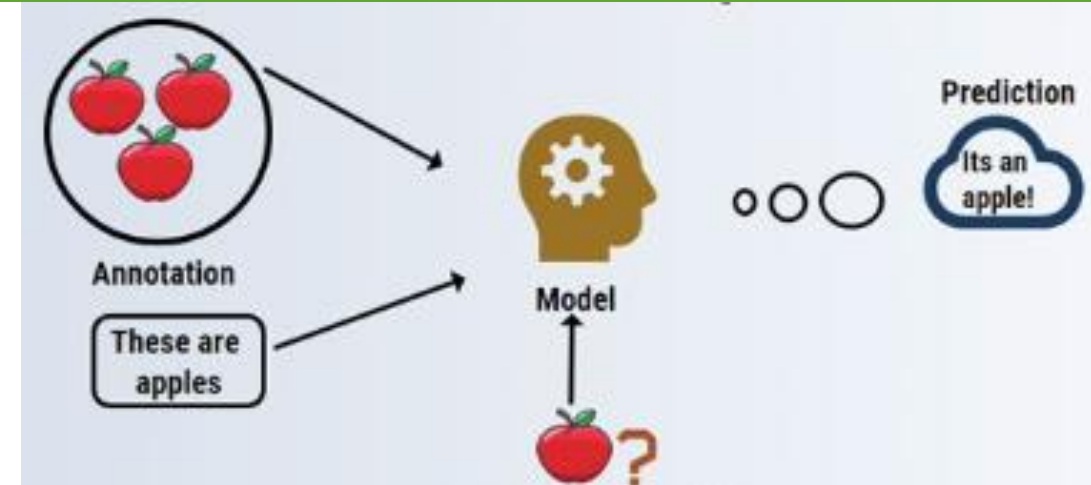
DL → When machines use **neural networks** to extract features
from raw data,
e.g: Models detecting forest fires



TYPES OF ML

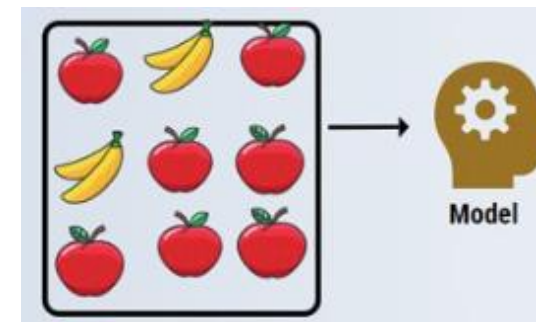
- **Supervised**

Models are trained on labelled data.



- **Unsupervised**

Models are trained on un-labelled data (without explicit output).

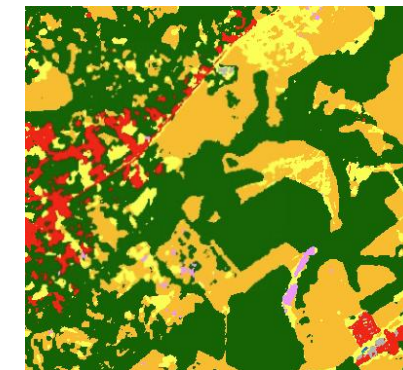
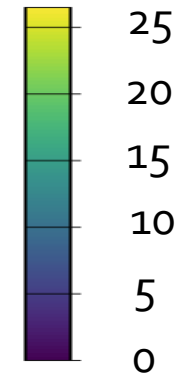
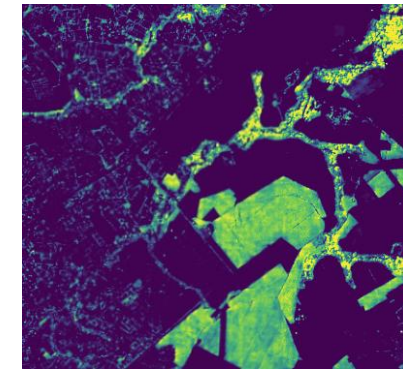
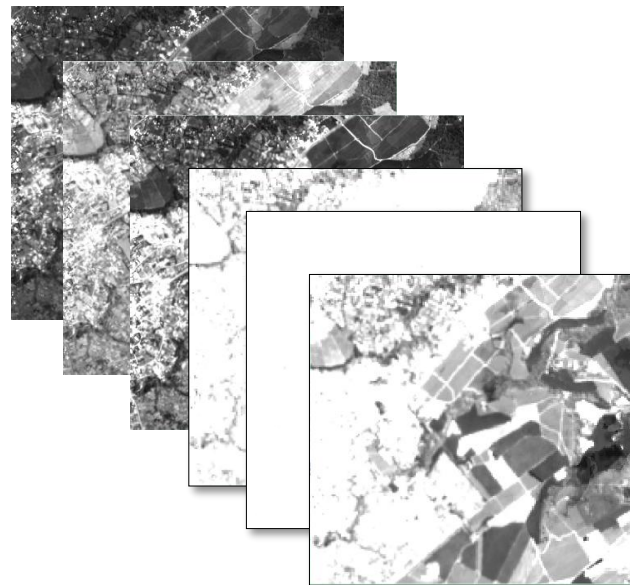


<https://shorturl.at/qkocG>

TURNING SATELLITE IMAGE INTO A VARIABLE

How do I turn a satellite image into a variable I can put on the right hand side of regression model?

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \dots + \beta_p X_{pi} + \varepsilon_i$$



LEARNING FROM EXAMPLES?

To learn from examples, we need examples.

Dataset curation forms the first step of any ML workflow.

The dataset comprises of:

- x:** input, predictors, features, independent (e.g. Sentinel-2 images, SRTM)
- y:** output, target, response, dependent (e.g. label, class, mask, bounding box)



Lon = 35.328749, Lat = -0.380411

CREATING LABELS

Class/Label	Landcover
1	Natural forest
2	Cropland
3	Planted forest
4	Built up
5	Water body



LABEL DATA

Typical complete model training dataset

Coordinates

Class

Features

lon	lat	label	B11	B12	B2	B3	B4	B5	B6	B7	B8	B8A	B9
35.26925	-0.4280921	4	0.10200	0.03540	0.01910	0.02960	0.01940	0.06700	0.24620	0.31050	0.30080	0.33480	0.32990
35.31435	-0.3824577	4	0.10320	0.03950	0.02290	0.03380	0.02210	0.06860	0.23680	0.29720	0.30460	0.32100	0.30500
35.32450	-0.3835357	3	0.03175	0.02035	0.04385	0.06595	0.04765	0.04075	0.03210	0.03615	0.04220	0.03625	0.08220
35.28075	-0.3536218	2	0.23240	0.21305	0.08100	0.10610	0.13340	0.15515	0.18495	0.19930	0.16325	0.21085	0.23085
35.28425	-0.4218039	4	0.10895	0.03930	0.01850	0.02785	0.01750	0.06435	0.25755	0.33095	0.33885	0.35460	0.36235
35.27204	-0.4262955	4	0.10070	0.03700	0.02040	0.03040	0.02030	0.06570	0.23200	0.29100	0.29800	0.31330	0.30720
35.30509	-0.3529032	2	0.30990	0.26610	0.10110	0.15150	0.26300	0.27430	0.32620	0.34020	0.36870	0.34480	0.36760
35.27904	-0.4110242	4	0.10680	0.04150	0.02030	0.02970	0.02030	0.06530	0.23480	0.29310	0.29470	0.31680	0.30510
35.32459	-0.3838950	3	0.02550	0.01660	0.04460	0.06520	0.04540	0.04190	0.03350	0.03900	0.03730	0.03350	0.08130
35.30105	-0.3585625	1	0.18760	0.07320	0.02990	0.07660	0.02950	0.14700	0.40850	0.50125	0.49365	0.52355	0.51410
35.30123	-0.3838052	4	0.12290	0.04430	0.02100	0.02950	0.01900	0.06830	0.28590	0.37220	0.38260	0.39820	0.39690
35.34453	-0.3724864	1	0.17745	0.07550	0.03040	0.07160	0.03230	0.13395	0.34685	0.40960	0.40885	0.43975	0.44070

DATASET NOTATION

$$D = \{(x^{(i)}, y^{(i)})\}_{(i=1)}^N$$

HOW TO GET LABEL DATA

Field Survey

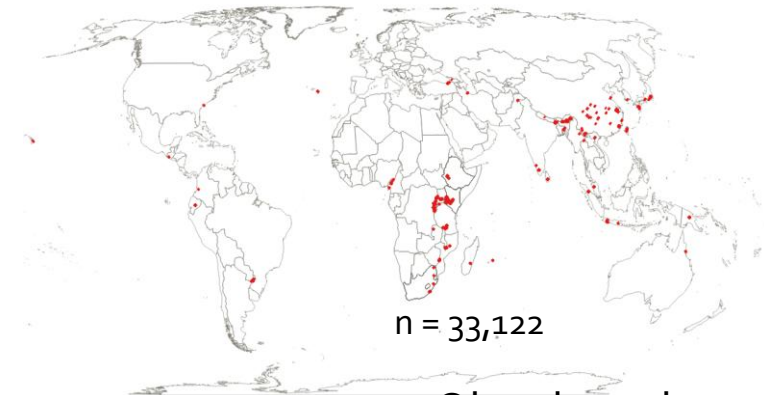
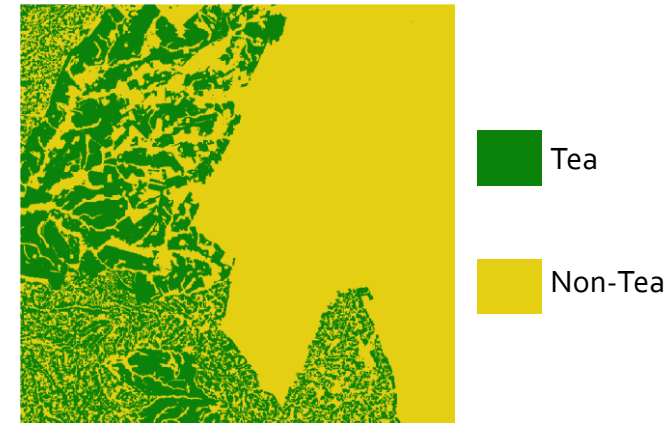


Manual Digitization



Google Earth Engine

Secondary sources



Oluoch et al., 2026

LABEL DATA PITFALLS

Field Survey

- Expensive
- Time consuming
- Limited spatial coverage
- Logistical constraints

Manual Digitization

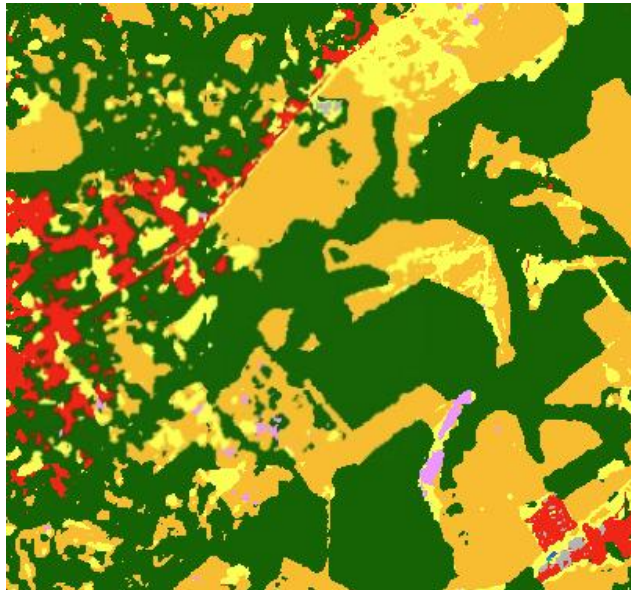
- Slow
- Subjective interpretation
- Basemap quality
- Difficulty at large scale

Secondary sources

- GIGO
- Resolution differences
- Temporal differences
- Unknown data quality
- Class differences

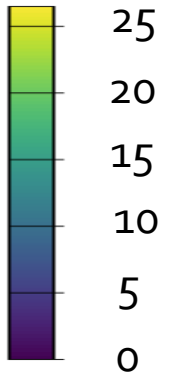
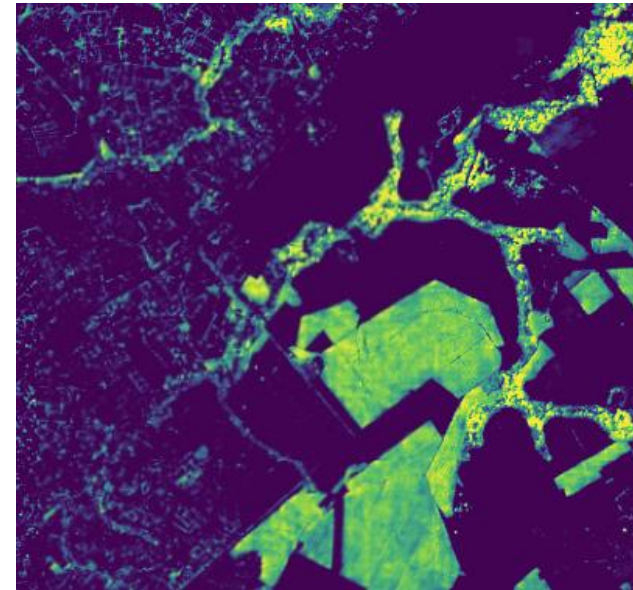
TYPICAL ML TASKS

Classification (Binary or Multi-class)



- Natural forest
- Cropland
- Sand quarry
- Built-up
- Solar PV

Regression



MODEL EVALUATION

Confusion matrix

y	$\hat{y}$	
		TP
		FN
		TP
		FP
		TP
		TN
		TP

	PP	PN
AP	TP	FN
AN	FP	TN

Accuracy

$$A = (TP + TN) / (TP + TN + FP + FN)$$

$$A = (4 + 1) / (4 + 1 + 1 + 1)$$

$$A = 0.71$$

Precision

$$P = (TP) / (TP + FP)$$

$$P = (4) / (4 + 1)$$

$$P = 0.80$$

Recall

$$R = (TP) / (TP + FN)$$

$$R = (4) / (4 + 1)$$

$$R = 0.80$$

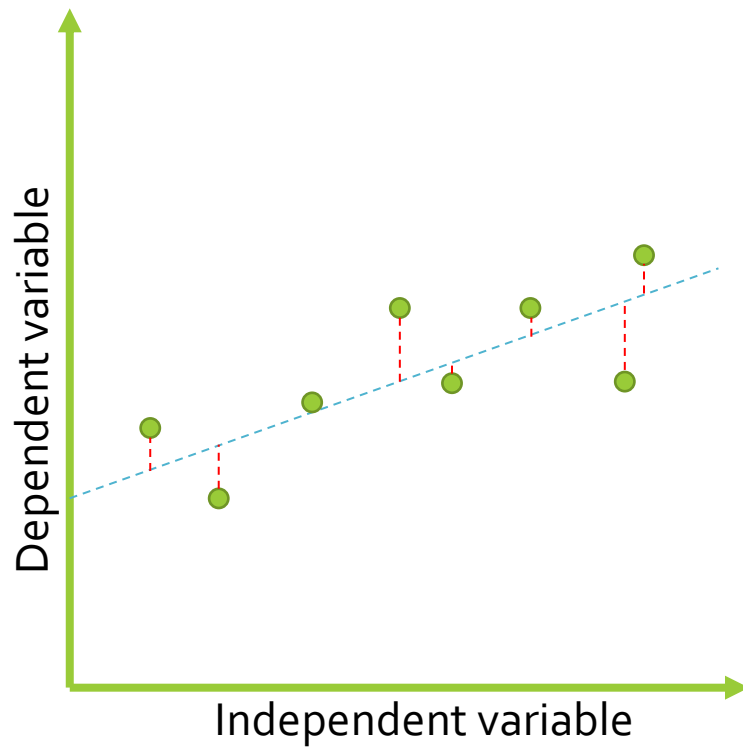
F1

$$F = 2 * \{(P * R) / (P + R)\}$$

$$F = 2 * \{(0.80 * 0.80) / (0.80 + 0.80)\}$$

$$F = 0.80$$

MODEL EVALUATION



y	$\hat{y}$	$ y - \hat{y} $
2	1.8	0.2
3	3.3	0.3
4	4.1	0.1
5	4	1.0
5.6	5.7	0.1
7	7.2	0.2
9	7.8	1.2
10	9.7	0.3

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

0.425

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

0.340

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2}$$

0.583

AREA OF APPLICABILITY



Kenya



Japan



India



Rwanda

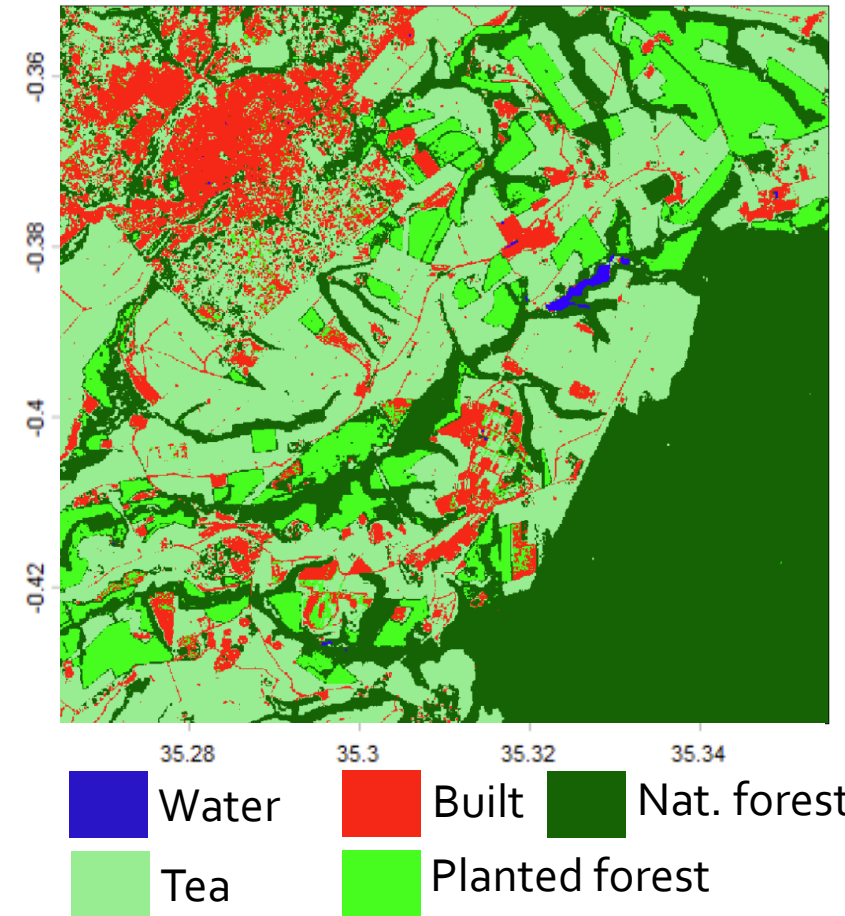
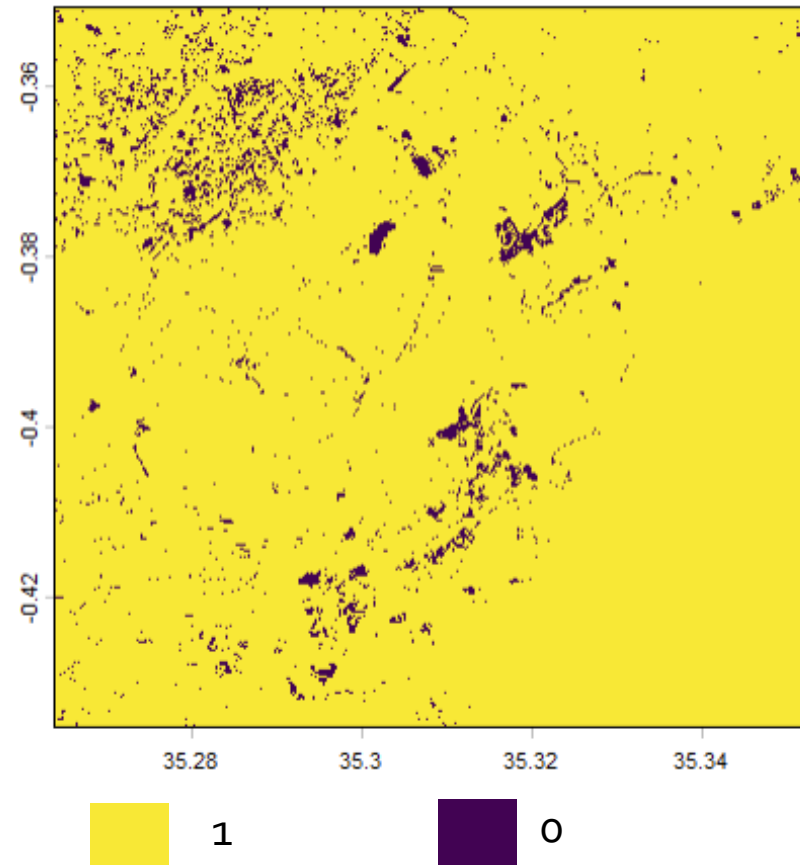
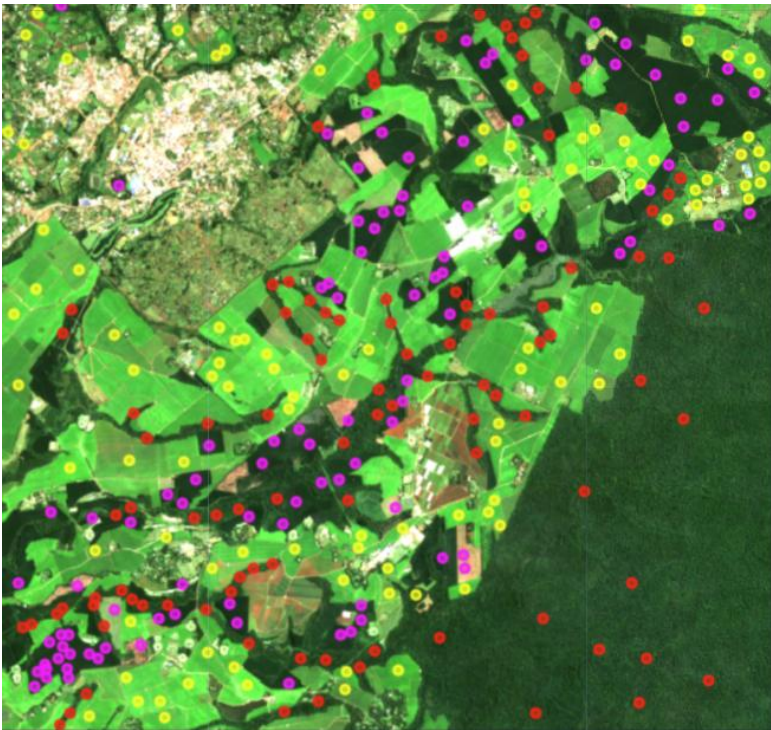


Kenya (cut-across pruning)

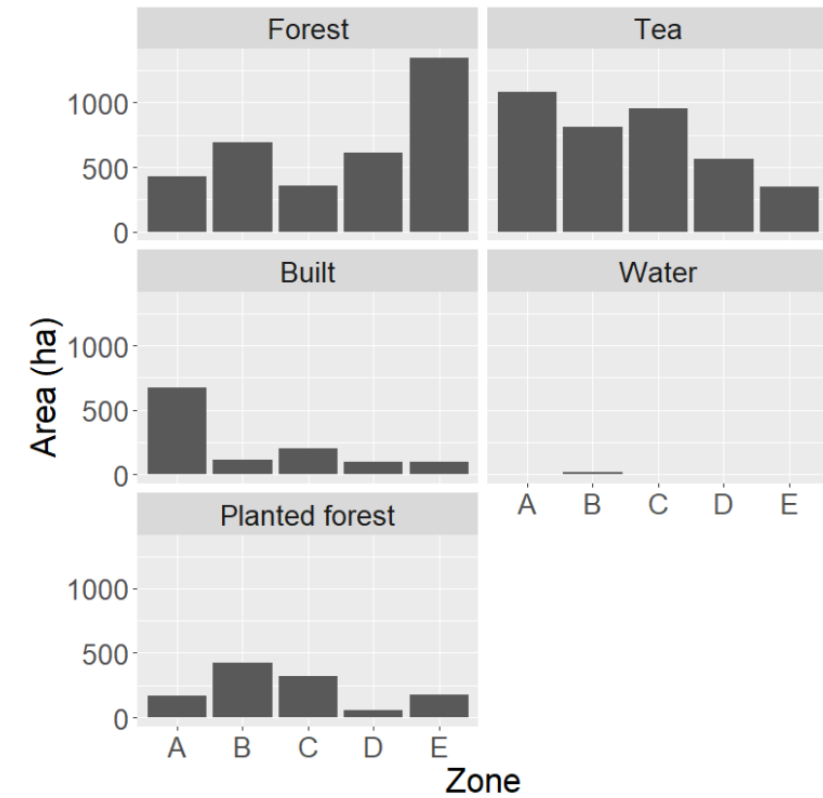
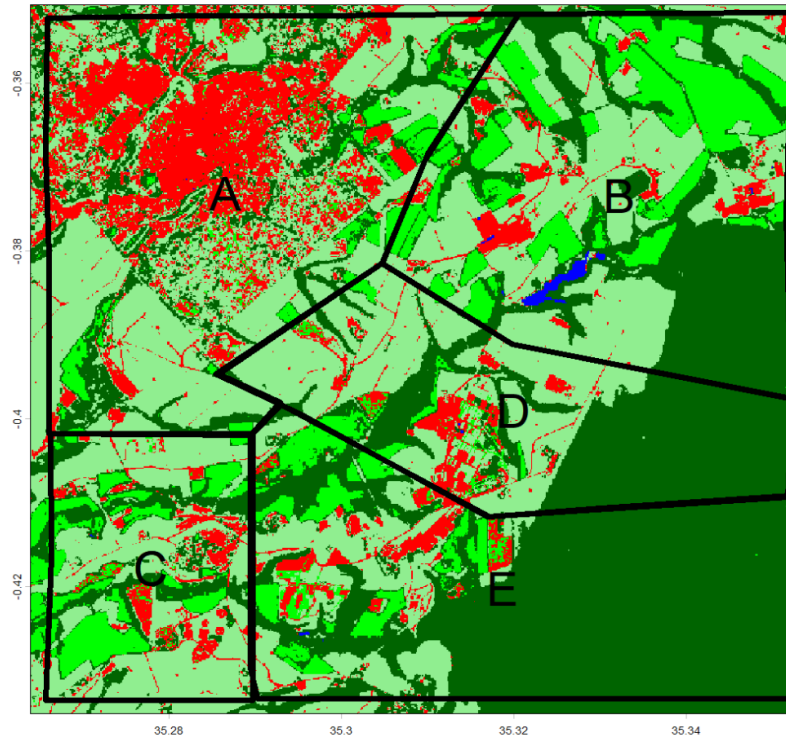
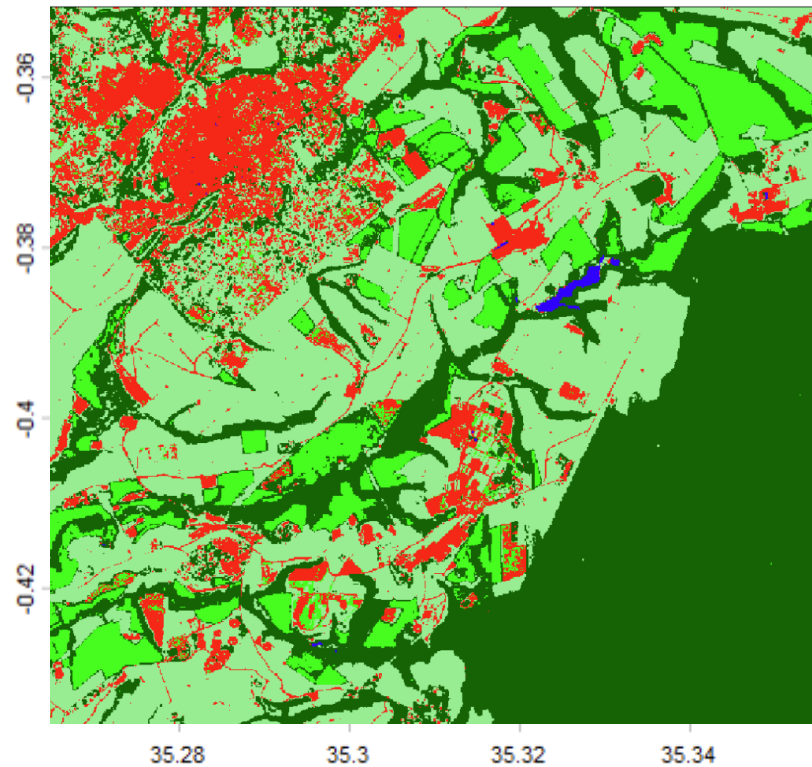


Kenya (lung-pruning)

AREA OF APPLICABILITY (MESS)



ZONAL STATISTICS



PANEL DATA

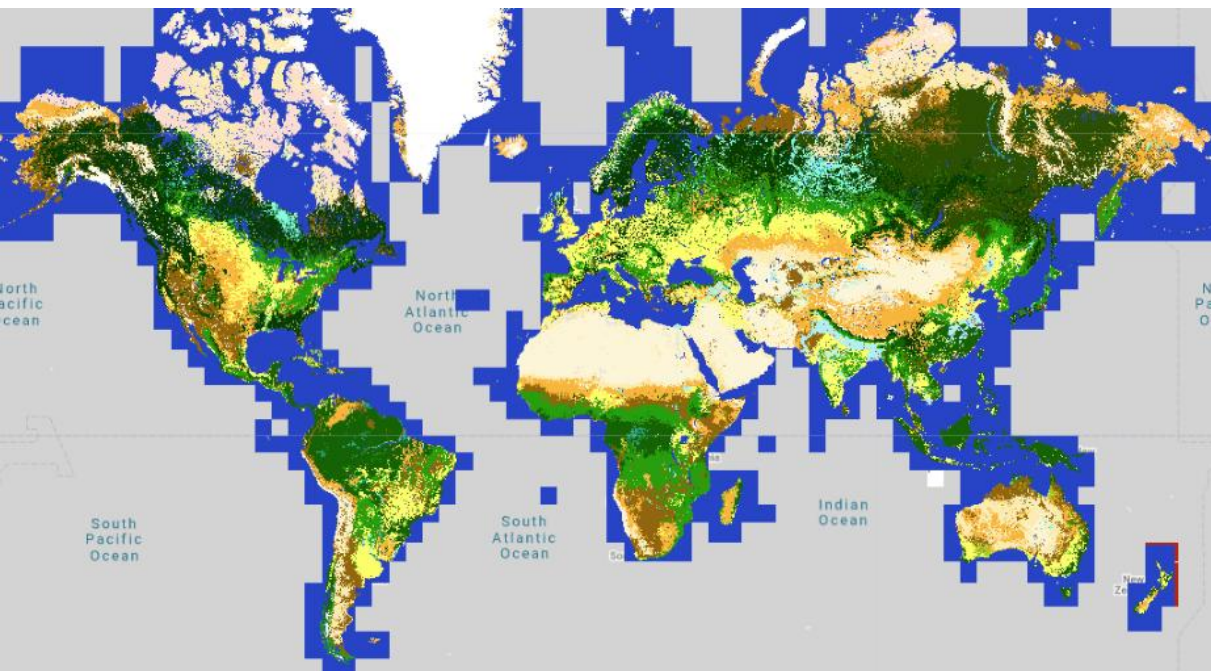
Zone	Year	Forest (ha)	Tea (ha)	Mean canopy height (m)
A	2023	440	1081	3.4
B	2023	703	804	7.1
C	2023	364	953	5.8
D	2023	616	563	8.6
E	2023	1354	346	13.2

POLICY RELEVANT QUESTIONS

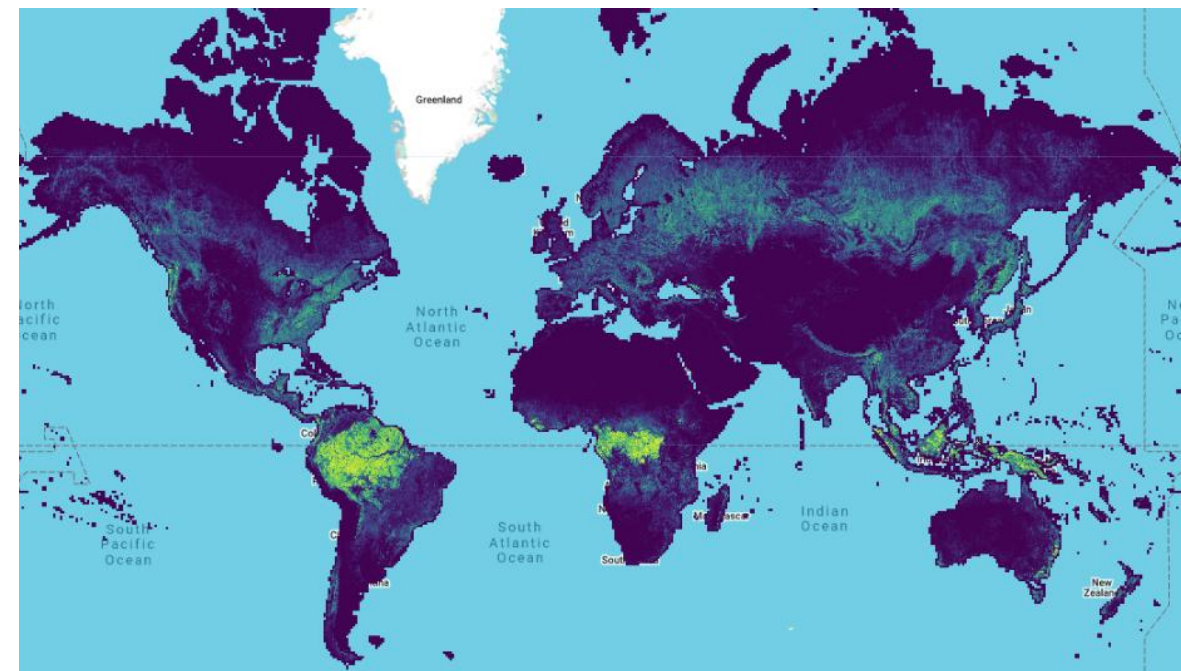
1. How much land per zone is under forest, and how did planted forest area change between x years?
2. Within the study years, tea cultivation expanded into which landcover?
3. Which zone has the fastest rate of forest loss and which one has highest landscape heterogeneity?
4. How does landscape simplification affect crop production per zone?
5. For zones where environmental policies were introduced in a given year, how did forest cover change per zone before versus after that year?

SAMPLE GLOBAL PRODUCTS

**GLC_FCS30D Global 30-meter Land Cover
Change Dataset (1985-2022)**



**Meta Global Canopy Height for 2018
- 2020 at 1 m resolution**



https://gee-community-catalog.org/projects/meta_trees/

SOLAR PV EXAMPLE

An electricity intensive factory installing solar pv panels in phases over time.



2019



2020



2024



2026

Lon: 36.695282° Lat: -1.087473°

LAND ECONOMICS GROUP



THANK YOU